

Problem 1: Decision tree performance at different depths

Depth	Precision	Recall	F1-Score
1	0.483333	0.666667	0.540230
2	0.972222	0.962963	0.965899
3	1.000000	1.000000	1.000000
4	1.000000	1.000000	1.000000
5	1.000000	1.000000	1.000000

1. The depth of the highest Recall

At depths 3, 4, and 5, Recall reaches 1.0000, which means that all positive examples (true category 1) are correctly classified.

Depth 2 has a recall of 0.9629, which means that some positives have been misclassified.

Conclusion: Starting max_depth=3, the model is already able to fully distinguish between categories without increasing depth.

2. Depth corresponding to minimum precision

At a depth of 1, Precision 0.4833 is the lowest, indicating a high percentage of false positives (many negatives are misclassified as positive).

As the depth increases, the Precision gradually increases, reaching 1.0000 at depth=3.

Conclusion: When the depth is too small (max_depth=1), the model has poor classification ability for negative cases.

3. Depth with the highest F1 score

depth=3 and above, with an F1 Score of 1.0000, indicates that the model has achieved the best balance between Precision and Recall.

Optimal depth: max_depth=3 because:

depth=3 has reached the perfect classification, and increasing the depth further will not improve performance.

Trees that are too deep may lead to overfitting, reducing generalization ability.

Micro Method

Calculation:

TP (True Case), FP (False Positive), FN (False Negative) for all categories are added together, and Precision, Recall, and F1 scores are calculated based on the total.

$$\text{Precision}_{micro} = \frac{\sum TP_i}{\sum (TP_i + FP_i)}$$

$$\text{Recall}_{micro} = \frac{\sum TP_i}{\sum (TP_i + FN_i)}$$

$$F1_{micro} = \frac{2 \cdot \text{Precision}_{micro} \cdot \text{Recall}_{micro}}{\text{Precision}_{micro} + \text{Recall}_{micro}}$$

Applicable Scenarios:

Ideal for balanced category distribution or when overall classification performance is a concern.

Advantages: Considering the overall classification situation, it is suitable for scenarios with a large number of categories.

Disadvantage: The effect of small categories can be masked when the categories are unevenly distributed.

Macro Averaging

Calculation:

Precision, recall, and F1 scores were calculated separately for each category, and then averaged across all categories.

$$\text{Precision}_{macro} = \frac{1}{N} \sum_{i=1}^N \text{Precision}_i$$

$$\text{Recall}_{macro} = \frac{1}{N} \sum_{i=1}^N \text{Recall}_i$$

$$\text{F1}_{macro} = \frac{1}{N} \sum_{i=1}^N \text{F1}_i$$

Applicable Scenarios:

Useful for focusing on the classification performance of each category, especially when the category distribution is uneven.

Advantages: Focus on the individual performance of each category and not be dominated by the majority.

Disadvantage: Overall performance may be underestimated when the number of categories is unbalanced.

Weighted Averaging

Calculation:

Precision, recall, and F1 scores were calculated separately for each category, and then weighted averaged based on the number of samples

in the test set for each category.

$$\text{Precision}_{weighted} = \frac{\sum_{i=1}^N w_i \cdot \text{Precision}_i}{\sum_{i=1}^N w_i}$$

$$\text{Recall}_{weighted} = \frac{\sum_{i=1}^N w_i \cdot \text{Recall}_i}{\sum_{i=1}^N w_i}$$

$$\text{F1}_{weighted} = \frac{\sum_{i=1}^N w_i \cdot \text{F1}_i}{\sum_{i=1}^N w_i}$$

Applicable Scenarios:

It is suitable for situations where the distribution of categories is uneven, and can reflect the impact of the proportion of categories in the data.

Advantages: Considers category imbalances and reflects overall classification performance.

Disadvantages: It may be dominated by categories with large amounts of data, resulting in the performance of small categories being neglected.

Problem 2: The first split in the decision tree

Selected features

First split characteristic: Uniformity_of_Cell_Size

Splitting threshold: 3.5

Index	total	left	right	weighted
Entropy	0.9165	0.3203	0.3014	0.3145
Gini	0.4432	0.1096	0.1014	0.1071
Misclassification Error	0.3315	0.0582	0.0536	-
Info Gain	0.6020	-	-	-

Problem 3: PCA impact analysis

Model 1: Using Only the First Principal Component

Confusion Matrix:

```
[[37  6]
```

```
 [ 0 71]]
```

False Positives (FP): 6

True Positives (TP): 71

False Positive Rate (FPR): 0.1395

True Positive Rate (TPR): 1.0000

Model 2: Using Both Principal Components

Confusion Matrix:

[[37 6]

[0 71]]

False Positives (FP): 6

True Positives (TP): 71

False Positive Rate (FPR): 0.1395

True Positive Rate (TPR): 1.0000

For the continuous Breast Cancer dataset, using PCA-reduced data (even just the first principal component) was as effective as using the original continuous data, indicating that PCA did not negatively impact the model's performance in this case.